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# Directamente conforme implica holomorfa

**Lema 1.** *Sea $f\colon \Omega \subset \mathbb{C} \to \mathbb{C}$ diferenciable en el sentido real y directamente conforme*

$$\Longrightarrow f \text{ es holomorfa en } \Omega.$$

***Demostración*:** Sea $(x, y) \in \Omega$, como $f$ es directamente conforme, $\exists \lambda > 0, Q \in SO(2)$ tales que $Df(x, y) = \lambda Q$ con $\det Q = 1$. Por tanto, $Q$ actúa como una rotación en $\mathbb{R}^2$, es decir, $\exists \theta \in \mathbb{R}$ tal que

$$Df(x, y) = \begin{pmatrix} \frac{\partial u}{\partial x}(x, y) & \frac{\partial u}{\partial y}(x, y) \\ \frac{\partial v}{\partial x}(x, y) & \frac{\partial v}{\partial y}(x, y) \end{pmatrix} = \lambda Q = \lambda \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

donde $u(x, y) = \Re(f(x + iy))$ y $v(x, y) = \Im(f(x + iy))$.

Observamos que se satisfacen las ecuaciones de Cauchy-Riemann en $(x, y)$:

$$\frac{\partial u}{\partial x}(x, y) = \lambda \cos\theta = \frac{\partial v}{\partial y}(x, y) \quad \wedge \quad \frac{\partial u}{\partial y}(x, y) = -\lambda \sin\theta = -\frac{\partial v}{\partial x}(x, y).$$

Dado que $f$ es $\mathbb{R}$-diferenciable, concluimos entonces que $f$ es $\mathbb{C}$-diferenciable en $(x, y)$ y, como este punto era arbitrario, tenemos que $f \in \mathcal{H}(\Omega)$. ■

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